

# Instruction-Guided Distribution Maximization for General Few-Shot Intent Recognition (Supplemental Materials)

## I. THEORETICAL ANALYSIS

**Proposition I.1.** *The distribution maximization approach enhances model generalization capabilities, and models optimized via IGDM exhibit lower upper bounds on generalization error over query samples compared to conventional empirical risk minimization (ERM) training methods.*

*Proof.* Standard ERM minimizes the average loss over the training dataset. In contrast, IGDM solves

$$\min_{\theta} \mathbb{E}_{(I, x, y_x) \sim D_{\text{train}}} \left[ \max_{E_x^T} \mathcal{L}(\theta; \text{Concat}(E_I, E_x^T), y_x) \right], \quad (1)$$

where the inner maximization is performed over embeddings  $E_x^T$  generated from the original query  $x$  by gradient ascent on the loss. This inner loop effectively searches for a worst-case perturbation of  $x$  within a neighborhood  $\mathcal{B}(E_x, \rho)$  in the embedding space, subject to task-consistent semantic constraints (Fig. 4). Consequently, Eq. (1) can be rewritten as the DRO surrogate

$$\mathcal{L}_{\text{igdm}}(\theta) = \frac{1}{|D_{\text{train}}|} \sum_{(I, x, y_x) \in D_{\text{train}}} \sup_{E_x^T \in \mathcal{B}(E_x, \rho)} \mathcal{L}(\theta; \text{Concat}(E_I, E_x^T), y_x). \quad (2)$$

This formulation aligns with the general DRO framework introduced by Ben-Tal et al. [1] and empirically studied in adversarial robustness [2], [3]. Unlike standard adversarial training that perturbs inputs in the token space, IGDM operates purely on the embedding representation, making it more efficient and semantically stable.

Let  $W(\cdot, \cdot)$  denote the Wasserstein distance between two distributions, and let

$$\mathcal{P}^{\rho} = \{D' : W(D', D_{\text{train}}) \leq \rho\} \quad (3)$$

be the uncertainty set of all probability distributions  $D'$  whose distance to the empirical training distribution  $D_{\text{train}}$  is at most  $\rho$ . The DRO [2] guarantees that, for any  $\theta$  and with high probability over the draw of the training set,

$$\sup_{D' \in \mathcal{P}^{\rho}} \mathbb{E}_{(x, y_x) \sim D'} [\mathcal{L}(\theta; x, y_x)] \leq \mathcal{L}_{\text{igdm}}(\theta) + \delta(|D_{\text{train}}|), \quad (4)$$

where  $\delta(\cdot)$  is a diminishing complexity term that depends on the sample size and the Rademacher complexity of the model class [4]. Inequality (4) states that minimizing  $\mathcal{L}_{\text{igdm}}(\theta)$  effectively minimizes the worst-case expected loss for all distributions  $D'$  that are  $\rho$ -close to  $D_{\text{train}}$ .

By design, the inner maximization in IGDM expands the support of the original query samples:  $\text{supp}(D_x) \subseteq \text{supp}(D_x^T)$ . Hence, the effective training distribution produced by IGDM, denoted  $D_{\text{train}}^{\text{eff}}$ , has a broader support than  $D_{\text{train}}$ . For the typical distribution shifts encountered in few-shot intent recognition, it holds that the test distribution satisfies  $D_{\text{test}} \in \mathcal{P}^{\rho}(D_{\text{train}}^{\text{eff}})$ . Standard ERM, on the other hand, only

provides guarantees for the exact empirical distribution  $D_{\text{train}}$ , leaving it vulnerable to test-time shifts.

We now compare the generalization error bounds of ERM and IGDM via the PAC-Bayes framework [5]. PAC-Bayes theory characterizes the generalization error  $\varepsilon_{\text{gen}}$  of a stochastic classifier by the Kullback-Leibler divergence  $\text{KL}$  between a posterior distribution  $Q$  over parameters and a fixed prior  $P$ .

For ERM, the learned parameters correspond to a sharp posterior  $Q_{\text{erm}}$  concentrated around a sharp minimum. The generalization error satisfies, with probability at least  $1 - \delta$ ,

$$\varepsilon_{\text{gen}}^{\text{erm}} \leq C_{\text{PAC}} \sqrt{\frac{\text{KL}(Q_{\text{erm}} \| P) + \log(n/\delta)}{n}}, \quad (5)$$

where  $n = |D_{\text{train}}|$  is the training set size,  $\delta \in (0, 1)$  is the confidence parameter, and  $C_{\text{PAC}}$  is a universal constant.

IGDM, through its inner maximization, augments the training set with perturbed embeddings. This augmentation is equivalent to training on an expanded support that covers more of the input space. It is well established that such augmentation encourages the optimizer to converge to flat minima, i.e., regions in the parameter space where the loss varies slowly, whereas ERM often ends up in sharp minima [6], [7]. Flat minima correspond to posteriors  $Q_{\text{igdm}}$  that are more diffuse and closer to the prior  $P$ , i.e.,  $\text{KL}(Q_{\text{igdm}} \| P) \leq \text{KL}(Q_{\text{erm}} \| P)$ .

Plugging this relation into the PAC-Bayes bound yields

$$\begin{aligned} \varepsilon_{\text{gen}}^{\text{igdm}} &\leq C_{\text{PAC}} \sqrt{\frac{\text{KL}(Q_{\text{igdm}} \| P) + \log(n/\delta)}{n}} \\ &\leq C_{\text{PAC}} \sqrt{\frac{\text{KL}(Q_{\text{erm}} \| P) + \log(n/\delta)}{n}} \approx \varepsilon_{\text{gen}}^{\text{erm}} \end{aligned} \quad (6)$$

Thus, IGDM provides a smaller or equal theoretical upper bound on the generalization error compared with conventional ERM training.  $\square$

**Proposition I.2.** *Distribution maximization only for query sample  $x$  is better than distribution maximization for instruction  $I$  simultaneously.*

*Proof.* Let the impact of embedding optimization of instruction part  $I$  and query part  $x$  on the loss be  $\Delta \mathcal{L}_I$  and  $\Delta \mathcal{L}_x$ , respectively.

We analyze the gradient variance by considering the underlying token-level structure. In Transformer-based language models, the embedding of a textual input is obtained by concatenating or summing the embeddings of its constituent tokens. Let the instruction  $I$  consist of  $m$  tokens, and the query  $x$  consist of  $l$  tokens. Their embeddings are  $E_I \in \mathbb{R}^{m \times d}$  and  $E_x \in \mathbb{R}^{l \times d}$ , where  $d$  is the hidden dimension. The gradient of the loss with respect to these embeddings, denoted  $\nabla_{E_I} \mathcal{L}$

and  $\nabla_{E_x} \mathcal{L}$ , are tensors of the same shapes. For the purpose of variance analysis, we vectorize these gradients and consider their squared norms or component-wise variances.

Typically, the instruction contains multiple descriptive sentences (task description, intent space enumeration, support examples), so  $m$  is substantially larger than  $l$ . Under the mild assumption that the gradient components corresponding to different tokens are approximately independent and have comparable variances, the total variance of the vectorized gradient scales linearly with the number of tokens. Formally,

$$\text{Var}(\nabla_{E_I} \mathcal{L}) \approx m \cdot \sigma^2, \quad \text{Var}(\nabla_{E_x} \mathcal{L}) \approx l \cdot \sigma^2 \quad (7)$$

where  $\sigma^2$  denotes the average per-token gradient variance. Since  $m \gg l$ , we have

$$\text{Var}(\nabla_{E_I} \mathcal{L}) \gg \text{Var}(\nabla_{E_x} \mathcal{L}). \quad (8)$$

This variance disparity implies that perturbing the instruction embedding introduces far greater stochasticity into the optimization process, which can lead to unstable outer updates and convergence to sharper minima with poorer generalization, as supported by the flat-minima theory.

From an information-theoretic perspective, the instruction  $I$  encodes the task semantics and the conditional relationship  $P(y|I, x)$ . Perturbing  $E_I$  directly corrupts this conditional distribution, increasing the conditional entropy  $H(y|I, x)$  and reducing the model's discriminative power. By the data processing inequality, any transformation of  $I$  cannot increase the mutual information between  $I$  and the intent label  $y$ ; i.e.,  $I(y; \tilde{I}, x) \leq I(y; I, x)$ , where  $\tilde{I}$  denotes the perturbed instruction. Hence, the perturbed instruction provides less task-relevant information, degrading performance. In contrast, perturbing only  $E_x$  corresponds to data augmentation on the marginal distribution  $P(x)$ , which expands the support of the input space without altering the conditional link  $y|x$ . This is equivalent to increasing the diversity of training samples, which, according to PAC-Bayesian generalization bounds, leads to a tighter bound on the expected risk.

Moreover, from a decomposition of the optimization objective:

$$\begin{aligned} \min_{\theta} \mathbb{E}_{(I, x, y_x) \sim D_0} [\max_{E_x} \mathcal{L}(\theta; \text{Concat}(E_I, E_x), y_x)] \\ = \min_{\theta} \mathbb{E}_{(I, x, y_x) \sim D_0} [\mathbb{E}_{(x, y_x | I)} [\max_{E_x} \mathcal{L}]], \end{aligned} \quad (9)$$

the inner expectation  $\mathbb{E}_{(x, y_x | I)}$  conditions on the instruction, indicating that the instruction remains fixed during query perturbation. This aligns perfectly with the inference scenario, where the instruction (including task description, intent space, and support examples) is fixed, and only the query varies. Simultaneously maximizing over  $E_I$  would optimize for unrealistic situations where the instruction itself is perturbed at test time, wasting computational resources and potentially overfitting to spurious noise.

By Pareto optimality [8], which advocates allocating limited computational resources to the most critical component, focusing on query distribution maximization achieves a better trade-off between robustness and task fidelity. The variance analysis demonstrates that perturbing the instruction incurs

a much larger cost in terms of optimization stability and generalization, while providing no benefit during inference. Therefore, maximizing only  $E_x$  yields superior generalization and robustness compared to simultaneous maximization of both embeddings.

Furthermore, the central conclusion of the above analysis can be further substantiated by additional evidence that extends beyond the gradient independence assumption. First, the information-theoretic argument already presented, which shows that perturbing  $I$  directly corrupts the task definition and increases  $H(y|I, x)$ , is entirely self-contained and provides sufficient grounds for keeping  $E_I$  fixed. Second, from an optimization geometry perspective, our inner-loop gradient ascent is implicitly bounded. Under the Lipschitz continuity of the loss landscape, the unconstrained inner maximization explores only a narrow, semantically-safe tubular neighborhood around the original embedding  $E_x$ . This is empirically verified in the Supplemental Materials, where (Fig. 4) we show that the cosine similarity between  $E_x$  and the perturbed  $E_x^T$  remains above 0.99, and the  $L_2$  norm difference is consistently on the order of  $10^{-2}$  throughout training. These observations confirm that the perturbation magnitude is naturally constrained without explicit norm clipping, mitigating the risk of semantic drift or instability. The combination of the information-theoretic constraint, the implicit regularization from the loss landscape, and the empirical evidence on semantic preservation collectively reinforce the proposition that confining distribution maximization to the query embedding alone is both theoretically sound and empirically optimal.  $\square$

## II. DATASETS

To comprehensively evaluate the method proposed in this paper, we investigate public datasets widely used in intent recognition in recent years and compile 20 single-label and 7 multi-label intent recognition datasets. The original sources, domain distributions, and scale information of all datasets are listed as follows:

- Single-label datasets (20): ACID [9], ATIS [10], Bank77 [11], BCS<sup>1</sup>, Clinic150 [12], Facebook [13], Hint3 [14], Hwu64 [15], LEYZER [16], Liu57 [15], MCID [17], MTOP [18], RedWood [19], SLURP [20], SNIPS [21], StackE [22], StackO [23], TOP [24], TOPv2 [25], XSID [26].
- Multi-label datasets (7): BlendATIS [27], BlendBank77 [27], BlendClinic150 [27], BlendSNIPS [27], NLU++ [28], Stanford [29], Toursg [30].

In the experiments, we split all datasets into general datasets and target datasets. The core purpose of this split is to simulate the real-world challenge of transferring knowledge from a limited general domain to novel domain tasks, and to validate the generalization ability of the proposed Instruction-Guided Distribution Maximization (IGDM) method without fine-tuning on target data. General datasets are used for instruction learning, corresponding to the pre-training stage. For single-label tasks, we select 5 datasets as general data: ACID, BCS,

<sup>1</sup><https://www.kaggle.com/datasets/scodepy/customer-support-intent-dataset>

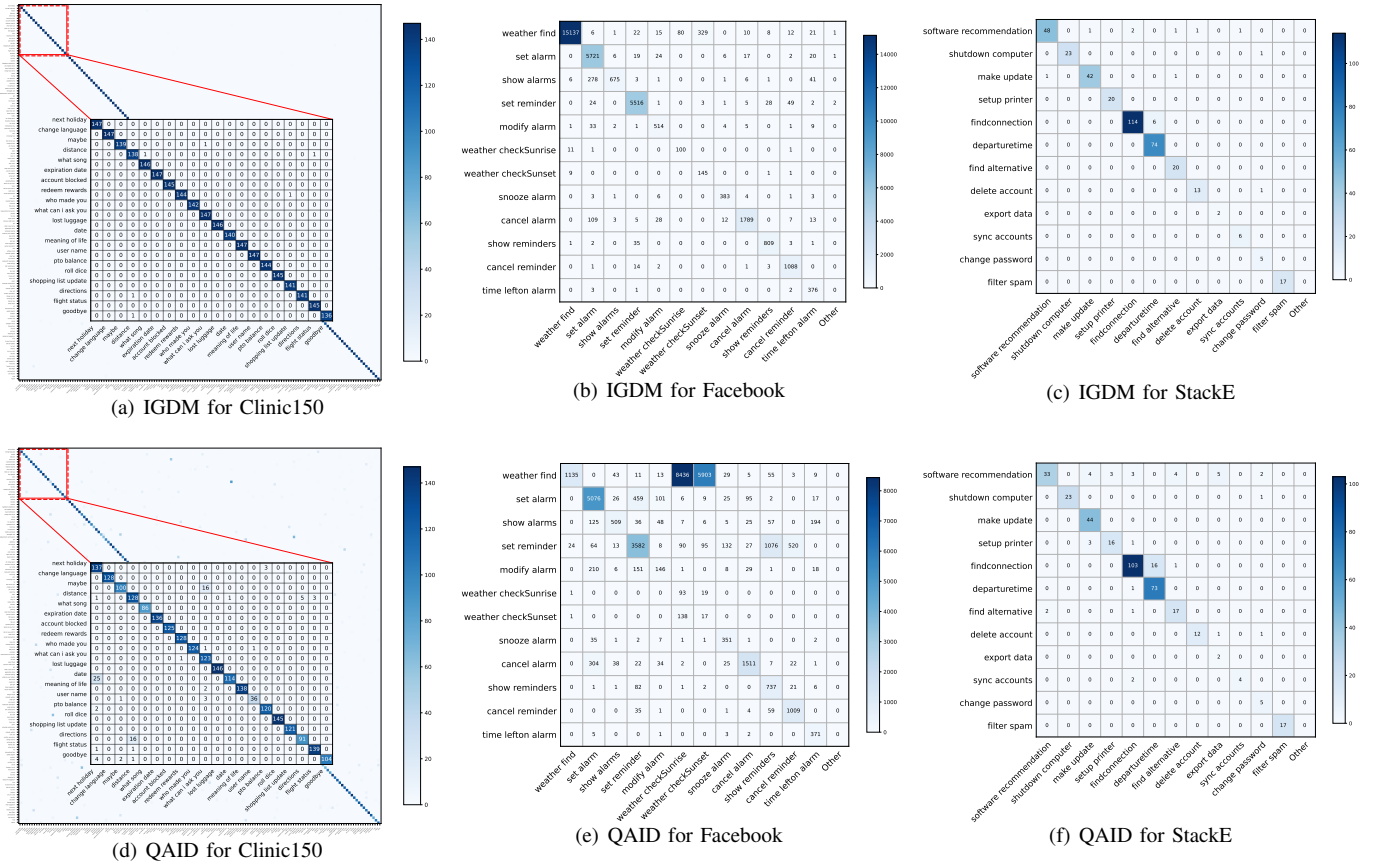


Fig. 1: Confusion matrix. The horizontal axis represents the predicted class and the vertical axis is the true class. Darker diagonals and larger numbers indicate better classification.

HINT3, MCID, XSID. For multi-label tasks, only BlendATIS is chosen as general data. Target datasets are used to evaluate the model’s zero-shot/few-shot generalization ability. For single-label tasks, the remaining 15 datasets (ATIS, Bank77, Clinic150, Facebook, Hwu64, Leyzer, Liu57, MTOP, Red-Wood, SLURP, SNIPS, StackE, StackO, TOP, TOPv2) serve as target data. For multi-label tasks, the remaining 6 datasets (BlendBank77, BlendClinic150, BlendSNIPS, NLU++, Stanford, Toursg) serve as target data.

To ensure no intent overlap or sample leakage between general and target datasets, we conduct strict data cleaning. For each general dataset, we manually inspect its intent labels and perform semantic comparison with intent labels in the target datasets, removing all categories and corresponding samples that are potentially similar to target intents. This step ensures that the knowledge learned by the model on general data does not directly come from target task categories, thus truly reflecting the model’s generalization ability when facing novel intents. In the target dataset evaluation phase, we follow the Any-way Any-shot setting: only  $K$  labeled samples per intent category are provided as support cases, which are fed into the model together with task instructions. The support cases are used for in-context learning and do not participate in model parameter updating. The remaining samples in the target dataset form the test set, used to compute evaluation metrics including accuracy and F1-score. For multi-label tasks, outputs

are formatted as strings in the form: “*Intent1, Intent2*”.

#### A. Predictive Behavior Analysis

In order to further explore the essential differences in the prediction behavior of different methods, this section will explore the confounding of model prediction in depth. We conducted a 2-shot comparison between IGDM and QAID on the three datasets Clinic150, Facebook, and StackE, and the prediction confusion matrix is shown in Fig. 1. Firstly, on clinic150, it can be clearly observed from Fig. 1(a) that the confusion matrix of IGDM appears almost as a pure diagonal line. A high concentration of values in the matrix means that the vast majority of examples are correctly classified, and the off-diagonal elements are near zero. This indicates that IGDM successfully summarizes highly refined decision boundaries between intents in this dataset through supporting cases, and result in few misjudgments. In contrast, the confusion matrix for QAID in Fig. 1(d) is significantly noisy and dispersed, with a large number of off-diagonal elements, indicating that significant inter-class confusion has occurred. For example, a large number of “data” are predicted as “next holiday”, and a large number of “directions” are predicted as “distance”. These errors indicate that QAID is more susceptible to interference from surface lexical similarity or shallow semantic associations, and it is difficult to learn sufficiently robust and precise decision boundaries. Similarly, on the Facebook

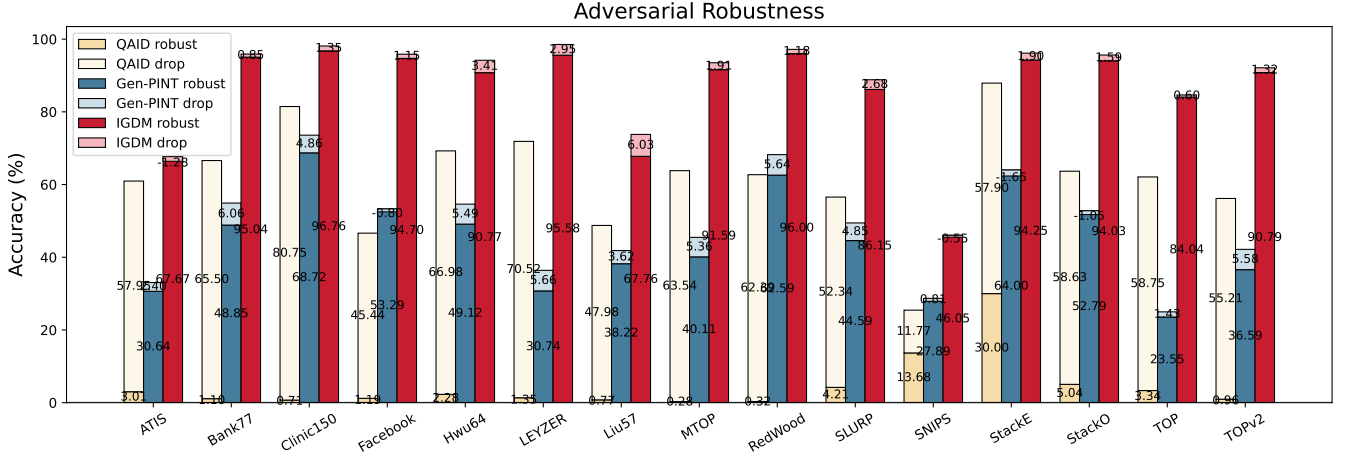


Fig. 2: Robust accuracy (%) for single-label few-shot intent recognition. IGDM robust represents the result after adversarial, and IGDM drop represents the result of adversarial drop.

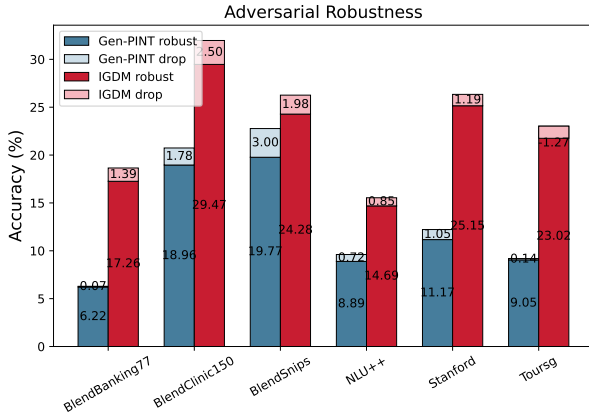


Fig. 3: Robust accuracy (%) for multi-label few-shot intent recognition.

dataset, QAID (Fig. 1(e)) heavily confuses "weather find" with other weather-related intents such as "weather checkSunrise/weather checkSunset". In contrast, IGDM (Fig. 1(b)) is able to avoid this kind of error relatively well. On the StackE dataset, the IGDM (Fig. 1(c)) matrix is very clear and the diagonal elements are absolutely dominant, which shows an overall high classification accuracy. Again, QAID (Fig. 1(f)) has obvious errors, such as a significant confusion between "findconnection" and "departuretime." Based on the above analysis, we know that IGDM can effectively resist the inter-class interference and produce higher prediction confidence. The QAID method relies heavily on direct similarity computation in the semantic embedding space (semantic retrieval) and is vulnerable to semantic misdirection (as Facebook Fig. 1(e)), which results in mismatching the input to a neighboring but wrong label. It is worth mentioning that QAID needs to be fine-tuned by a small number of samples on the target data, but QAID is not satisfactory in the results. In contrast, IGDM does not need to update parameters when generalizing the target data, but uses the instruction formulation to complete the generalization. IGDM is able to learn decision boundaries

with better quality than the baseline QAID, which makes it a more stable performing intent recognition scheme.

### B. Robust Analysis

To further verify the robustness and generalization ability of the model, we perform sample attacks on both single-label and multi-label datasets. Specifically, we leverage a T5 paraphrasing model to generate paraphrasing text for the target data, which is then used to replace the original test data to obtain robustness results. As shown in Fig. 2 and Fig. 3, in the test of adversarial example attack, the proposed IGDM shows significantly better robustness than the baseline model on both single-label and multi-label few-shot intent recognition. In the single-label task (Fig. 2), IGDM maintains high robustness results on each dataset, which are significantly higher than QAID and Gen-PINT (for example, on Bank77 after the attack, IGDM is 95.04%, while QAID is 48.85% and Gen-PINT is only 1.10%). The average performance degradation of IGDM is only 1.67%, which is far lower than 57.04% of QAID and 3.22% of Gen-PINT. It is particularly worth noting that on ATIS and SNIPS datasets, the accuracy of IGDM after attack increases by 1.28% and 0.55% respectively, which indicates that IGDM may discriminate some adversarial samples more accurately, and reflects the superiority of its decision boundary. On the contrary, QAID can obtain the second result in most cases before the attack, but its performance decreases greatly after the attack, which indicates that its robustness and generalization ability are low. In the multi-label task (Fig. 3), although the absolute accuracy of all models generally decreases due to the increase of task complexity, IGDM still maintains its relative advantage, and its average robustness result (21.75%) is more than twice that of Gen-PINT(9.19%). The average performance degradation after attack (1.11%) is less than that of Gen-PINT(1.13%), and it shows better tolerance on BlendClinic150 and Stanford data sets. Taken together, IGDM explicitly expands the tolerance range of the model to errors through the min-max optimization strategy, so



TABLE I: Distribution maximization combined with other base models on single-label accuracy (%).

| Method          | ATIS         | Bank77       | Clinic150    | Facebook     | Hwu64        | LEYZER       | Liu57        | MTOP         |
|-----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| BERT 1-shot     | 6.72         | 19.48        | 36.72        | 48.13        | 24.40        | 55.72        | 21.48        | 28.22        |
| BERT 2-shot     | 28.07        | 43.61        | 67.02        | 62.98        | 52.87        | 68.32        | 37.65        | 51.10        |
| BERT+DM 1-shot  | 38.80        | 33.47        | 47.80        | 53.39        | 39.04        | 60.15        | 30.52        | 31.62        |
| BERT+DM 2-shot  | <b>47.01</b> | <b>50.46</b> | <b>66.89</b> | <b>67.92</b> | <b>55.30</b> | <b>71.64</b> | <b>46.14</b> | <b>54.36</b> |
| GPT-2 0-shot    | 0.17         | 3.16         | 4.02         | 0.00         | 2.67         | 1.23         | 0.59         | 2.05         |
| GPT-2 1-shot    | 0.15         | 13.62        | 9.10         | 1.10         | 6.70         | 7.07         | 5.91         | 6.24         |
| GPT-2 2-shot    | 0.19         | 11.90        | 9.30         | 1.52         | 30.28        | <b>24.69</b> | 3.74         | 7.75         |
| GPT-2+DM 0-shot | 0.14         | 5.23         | 1.66         | 0.02         | 3.23         | 3.43         | 1.20         | 4.84         |
| GPT-2+DM 1-shot | <b>11.57</b> | <b>29.81</b> | 19.82        | 35.60        | 41.93        | 10.92        | <b>10.15</b> | <b>19.79</b> |
| GPT-2+DM 2-shot | 0.17         | 29.11        | <b>27.28</b> | <b>35.97</b> | <b>45.26</b> | 10.11        | 5.27         | 17.80        |
| Method          | RedWood      | SLURP        | SNIPS        | StackE       | StackO       | TOP          | TOPv2        | Avg.         |
| BERT 1-shot     | 25.23        | 19.59        | 13.84        | 51.70        | 5.82         | 26.61        | 29.05        | 27.51        |
| BERT 2-shot     | 52.55        | 36.65        | 17.12        | 61.65        | 23.70        | 37.47        | 46.72        | 45.83        |
| BERT+DM 1-shot  | 33.87        | 25.19        | 17.97        | 67.50        | 11.70        | 43.34        | 32.58        | 37.80        |
| BERT+DM 2-shot  | <b>61.23</b> | <b>40.18</b> | <b>24.60</b> | <b>71.50</b> | <b>33.85</b> | <b>57.46</b> | <b>51.08</b> | <b>53.31</b> |
| GPT-2 0-shot    | 4.89         | 4.04         | 0.29         | 0.25         | 0.88         | 0.17         | 1.55         | 1.73         |
| GPT-2 1-shot    | 7.44         | 12.01        | 20.76        | 20.50        | 9.20         | 0.47         | 13.25        | 8.90         |
| GPT-2 2-shot    | 9.08         | 11.23        | <b>22.45</b> | <b>36.25</b> | 11.66        | 1.90         | 8.58         | 12.70        |
| GPT-2+DM 0-shot | 3.77         | 5.41         | 2.12         | 0.75         | 4.21         | 0.17         | 4.63         | 2.72         |
| GPT-2+DM 1-shot | 7.04         | <b>26.39</b> | 12.28        | 35.50        | 69.81        | <b>40.29</b> | 18.58        | 25.97        |
| GPT-2+DM 2-shot | <b>26.05</b> | 9.05         | 8.53         | 31.00        | <b>72.66</b> | 34.40        | <b>43.02</b> | <b>26.38</b> |

that it can maintain high performance and high robustness in different types of tasks.

### III. DISTRIBUTION MAXIMIZATION WITH DIFFERENT BASE MODELS

To further verify the effectiveness of distribution maximization, we apply it to the BERT and GPT-2 models, with the experimental results shown in Tables I, II, III and IV. The results demonstrate that DM can significantly and consistently improve the performance of different base models on both single-label and multi-label few-shot intent recognition tasks. In single-label tasks, the average accuracy of BERT+DM under 1-shot and 2-shot settings is improved by more than 10% and 7% respectively compared with vanilla BERT (e.g., from 6.72% to 38.80% on the ATIS dataset). Even for GPT-2, which has inherently weak few-shot capabilities by default, after introducing DM, its 1-shot average accuracy rises from approximately 15% to over 26%. In the more challenging multi-label setting, the gains brought by DM are generally significant. For instance, BERT+DM substantially outperforms the baselines in accuracy and F1-score across multiple datasets. Notably, on the Stanford dataset, the 2-shot accuracy jumps from 0% to 2.69%.

The core reason why the DM strategy yields consistent and stable performance improvements is that it generates “hard samples” in the embedding space of query instances via gradient ascent, proactively exploring and expanding the high-uncertainty regions near the current model’s decision boundary. This is equivalent to simulating potential distribution shifts during training, forcing the model to learn more robust and discriminative features under limited data, thereby effectively alleviating overfitting. For discriminative models such as BERT, DM helps refine class boundaries; for generative models such as GPT-2, DM enhances its tolerance to

input perturbations. This indicates that, as a generalization-enhancing and robustness-improving tool, DM can effectively boost the performance of various base models including BERT, GPT-2, and Flan-T5.

### IV. SEMANTIC SIMILARITY DURING DISTRIBUTION MAXIMIZATION

During the training of IGDM, we further investigate the impact of distribution maximization on sample semantics. Specifically, we record the cosine similarity between the embedding  $E_x$  of a query sample before distribution maximization (DM) and its representation  $E_x^T$  after DM, and track the trend of this similarity across training epochs. To provide a more complete assessment, we simultaneously compute the absolute L2 norm difference between the original and perturbed embeddings for every sample undergoing distribution maximization. Both metrics are aggregated over all mini-batches within each epoch and averaged to obtain epoch-level trends. The experimental results are shown in Fig. 4, where the left displays the cosine similarity curve and the right presents the L2 norm difference curve. The similarity remains above 0.99 throughout the entire training process without any noticeable decline, while the L2 norm difference is consistently maintained at an extremely low magnitude (on the order of  $10^{-2}$  or below) across all training epochs. This phenomenon indicates that although IGDM actively amplifies the model’s prediction loss in the embedding space via gradient ascent, the introduced perturbation does not alter the core semantic information of the sample, nor does it cause uncontrolled scaling or collapse of the embedding vectors.

The main reason is that even though the inner-loop optimization amplifies the loss through gradient ascent, pushing the sample toward high-uncertainty regions near the decision boundary, such perturbation is strictly constrained within the

TABLE II: Distribution maximization combined with other base models on multi-label F1-score (%).

| Method          | ATIS         | Bank77       | Clinic150    | Facebook     | Hwu64        | LEYZER       | Liu57        | MTOP         |
|-----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| BERT 1-shot     | 9.40         | 18.96        | 35.47        | 31.76        | 22.63        | 50.53        | 19.79        | 17.86        |
| BERT 2-shot     | 28.80        | 42.82        | 65.63        | 50.69        | 50.84        | 67.79        | 38.50        | 33.06        |
| BERT+DM 1-shot  | 33.41        | 32.50        | 45.63        | 34.01        | 36.36        | 52.56        | 29.27        | 20.32        |
| BERT+DM 2-shot  | <b>49.36</b> | <b>49.93</b> | <b>65.97</b> | <b>59.45</b> | <b>53.10</b> | <b>70.55</b> | <b>44.12</b> | <b>38.07</b> |
| GPT-2 0-shot    | 0.41         | 0.85         | 1.34         | 0.00         | 0.42         | 0.78         | 0.12         | 0.46         |
| GPT-2 1-shot    | 0.45         | 0.94         | 2.27         | 0.04         | 0.21         | 0.94         | 0.45         | 0.63         |
| GPT-2 2-shot    | 0.56         | 0.95         | 1.93         | 0.03         | 3.30         | <b>9.95</b>  | 0.39         | 0.72         |
| GPT-2+DM 0-shot | 0.16         | 1.15         | 0.68         | 0.00         | 0.47         | 1.09         | 0.36         | 0.66         |
| GPT-2+DM 1-shot | <b>15.48</b> | 4.52         | 9.53         | 5.13         | <b>9.24</b>  | 3.62         | <b>2.82</b>  | 2.30         |
| GPT-2+DM 2-shot | 0.21         | <b>6.55</b>  | <b>15.04</b> | <b>5.74</b>  | 8.49         | 5.10         | 2.56         | <b>4.42</b>  |
| Method          | RedWood      | SLURP        | SNIPS        | StackE       | StackO       | TOP          | TOPv2        | Avg.         |
| BERT 1-shot     | 22.33        | 15.53        | 7.79         | 37.94        | 2.23         | 13.80        | 23.70        | 21.98        |
| BERT 2-shot     | 48.28        | 29.45        | 10.48        | 46.43        | 21.53        | 21.76        | 40.25        | 39.75        |
| BERT+DM 1-shot  | 30.46        | 19.28        | 16.46        | 53.35        | 10.89        | 22.23        | 27.46        | 30.95        |
| BERT+DM 2-shot  | <b>56.77</b> | <b>32.55</b> | <b>23.06</b> | <b>65.44</b> | <b>31.95</b> | <b>29.92</b> | <b>45.12</b> | <b>47.69</b> |
| GPT-2 0-shot    | 0.97         | 0.29         | 0.08         | 0.67         | 0.05         | 0.02         | 0.28         | 0.45         |
| GPT-2 1-shot    | 0.84         | 0.21         | 0.65         | 10.23        | 0.25         | 0.75         | 0.61         | 1.30         |
| GPT-2 2-shot    | 0.70         | 0.23         | 1.66         | 3.61         | 0.24         | <b>2.72</b>  | 0.34         | 1.82         |
| GPT-2+DM 0-shot | 1.01         | 0.39         | 0.23         | 1.57         | 0.13         | 0.01         | 0.43         | 2.82         |
| GPT-2+DM 1-shot | 4.88         | <b>5.83</b>  | <b>4.56</b>  | 13.31        | 3.29         | 0.83         | 0.52         | 3.82         |
| GPT-2+DM 2-shot | <b>9.55</b>  | 4.45         | 0.95         | <b>22.26</b> | <b>3.77</b>  | 0.98         | <b>1.92</b>  | <b>4.82</b>  |

TABLE III: Distribution maximization combined with other base models on multi-label accuracy (%).

| Method          | BlendB.     | BlendC.     | BlendS.     | NLU++       | Stanford    | Toursg      | Avg.        |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| BERT 1-shot     | 0.07        | 0.44        | 2.51        | 0.47        | 0.00        | <b>1.55</b> | 0.84        |
| BERT 2-shot     | 0.08        | 1.61        | 1.20        | 0.51        | 0.00        | 0.10        | 0.58        |
| BERT+DM 1-shot  | <b>1.17</b> | 1.14        | 2.28        | <b>0.82</b> | 0.16        | 1.20        | 1.13        |
| BERT+DM 2-shot  | 1.12        | <b>1.77</b> | <b>3.22</b> | 0.70        | <b>2.69</b> | 1.38        | <b>1.81</b> |
| GPT-2 1-shot    | 0.00        | 0.02        | 0.10        | 0.01        | 0.15        | 0.00        | 0.05        |
| GPT-2 2-shot    | 0.00        | 0.03        | 0.24        | 0.49        | 0.46        | 0.01        | 0.21        |
| GPT-2+DM 1-shot | <b>0.30</b> | 1.13        | 1.15        | <b>1.08</b> | 0.94        | 0.07        | 0.78        |
| GPT-2+DM 2-shot | 0.11        | <b>1.32</b> | <b>2.18</b> | 1.04        | <b>1.49</b> | <b>0.90</b> | <b>1.17</b> |

TABLE IV: Distribution maximization combined with other base models on multi-label F1-score (%).

| Method          | BlendB.     | BlendC.     | BlendS.     | NLU++       | Stanford    | Toursg      | Avg.        |
|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| BERT 1-shot     | 0.61        | <b>0.67</b> | 1.36        | 0.04        | 0.00        | <b>0.22</b> | 0.48        |
| BERT 2-shot     | 0.43        | 0.63        | 0.87        | 0.06        | 0.00        | 0.25        | 0.37        |
| BERT+DM 1-shot  | <b>1.08</b> | 0.64        | <b>1.55</b> | <b>0.91</b> | 0.41        | 0.53        | 0.85        |
| BERT+DM 2-shot  | <b>1.67</b> | <b>0.65</b> | <b>1.39</b> | 0.23        | <b>1.29</b> | <b>0.86</b> | <b>1.02</b> |
| GPT-2 1-shot    | 0.00        | 0.02        | 0.10        | 0.01        | 0.15        | 0.00        | 0.05        |
| GPT-2 2-shot    | 0.00        | 0.03        | 0.24        | <b>0.49</b> | 0.46        | <b>0.01</b> | 0.21        |
| GPT-2+DM 1-shot | <b>0.01</b> | <b>0.42</b> | <b>0.26</b> | <b>0.21</b> | 0.28        | 0.00        | 0.20        |
| GPT-2+DM 2-shot | <b>0.03</b> | <b>0.23</b> | <b>0.24</b> | 0.13        | <b>0.67</b> | <b>0.01</b> | <b>0.22</b> |

continuous feature space and does not change the discrete semantic expression of the sample. The lower L2 norm difference further corroborates that the perturbation magnitude remains well-bounded: the inner-loop gradient ascent explores only a narrow, semantically-safe tubular neighborhood around the original embedding, without inducing substantial vector expansion or contraction that could artifactually distort the model’s internal representations. This finding reveals that IGDM enhances model robustness not by introducing semantic

drift or noise, but by forcing the model to learn wider and smoother decision boundaries while preserving semantic consistency in both directional orientation and representation scale. Such a mechanism enables the model to still accurately capture the essence of user intent when facing distribution shifts during testing, thereby achieving superior generalization performance in few-shot intent recognition tasks.

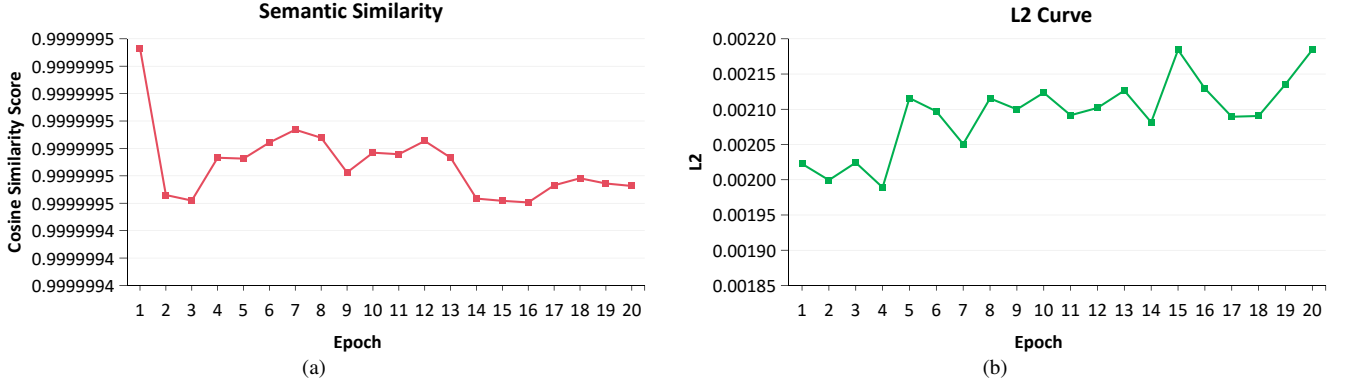


Fig. 4: Sample semantic similarity and L2 magnitude before and after distribution maximization. We can observe that the samples before and after distribution maximization have high semantic similarity, and this semantic similarity is maintained with the iteration of training rounds. In addition, the L2 norm difference is always maintained at an extremely low value. This dual perspective confirms that IGDM’s inner gradient ascent perturbations do not materially pull the embedding in the direction of magnitude inflation or collapse, and hardly change its semantic direction.

TABLE V: Multi-label accuracy (%) of BlendClinic150 as generic training data.

| Method            | BlendB.      | BlendA.      | BlendS.      | NLU++        | Stanford     | Toursg       | Avg.         |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| IntentBERT 1-shot | 1.14         | 1.01         | 1.79         | 1.06         | 2.05         | 0.07         | 1.19         |
| IntentBERT 2-shot | 1.25         | 1.33         | 1.44         | 1.62         | 1.43         | 0.05         | 1.19         |
| JSCL 1-shot       | 0.84         | 1.09         | 3.74         | 0.43         | 7.44         | 2.12         | 2.61         |
| JSCL 2-shot       | 0.90         | 1.53         | 4.07         | 0.71         | 15.29        | 5.78         | 4.71         |
| LaKG 1-shot       | 0.12         | 0.02         | 2.70         | 0.06         | 3.42         | 1.51         | 1.31         |
| LaKG 2-shot       | 0.09         | 0.01         | 6.03         | 0.18         | 1.05         | 1.38         | 1.46         |
| Flan-T5 1-shot    | 3.41         | 26.24        | 5.43         | 4.58         | 19.32        | 3.21         | 10.37        |
| Flan-T5 2-shot    | 7.53         | <b>43.38</b> | 7.87         | 7.35         | <b>24.28</b> | 7.34         | 16.29        |
| Gen-PINT 0-shot   | 6.27         | 20.09        | <b>28.70</b> | 11.15        | 10.91        | 8.02         | 14.19        |
| Gen-PINT 1-shot   | 7.58         | 40.95        | 23.26        | 13.28        | 12.87        | 9.23         | 17.86        |
| Gen-PINT 2-shot   | 7.13         | 40.20        | <b>26.58</b> | 13.56        | 12.48        | 9.49         | <b>18.24</b> |
| IGDM 0-shot       | <b>10.29</b> | 28.53        | 20.22        | <b>15.20</b> | 11.04        | <b>14.36</b> | 16.61        |
| IGDM 1-shot       | <b>20.67</b> | <b>41.51</b> | <b>23.74</b> | <b>21.03</b> | <b>21.49</b> | <b>19.73</b> | <b>24.70</b> |
| IGDM 2-shot       | <b>22.56</b> | <b>43.69</b> | 23.32        | <b>21.34</b> | <b>24.87</b> | <b>27.12</b> | <b>27.15</b> |

TABLE VI: Multi-label F1-score (%) of BlendClinic150 as generic training data.

| Method            | BlendB.     | BlendA.      | BlendS.     | NLU++       | Stanford     | Toursg      | Avg.        |
|-------------------|-------------|--------------|-------------|-------------|--------------|-------------|-------------|
| IntentBERT 1-shot | <b>1.94</b> | 1.23         | 2.35        | <b>1.47</b> | 1.06         | 0.13        | 1.36        |
| IntentBERT 2-shot | <b>2.67</b> | 1.07         | 1.88        | <b>2.20</b> | 2.61         | 0.11        | 1.76        |
| JSCL 1-shot       | 0.14        | 0.59         | <b>4.39</b> | 0.40        | <b>5.09</b>  | <b>1.53</b> | 2.02        |
| JSCL 2-shot       | 0.93        | 1.06         | <b>3.75</b> | 0.77        | <b>12.11</b> | <b>4.26</b> | <b>3.81</b> |
| LaKG 1-shot       | 1.01        | 0.64         | 2.44        | 0.16        | 2.78         | 0.72        | 1.29        |
| LaKG 2-shot       | 0.72        | 0.66         | <b>5.13</b> | 0.64        | 2.02         | 0.77        | 1.66        |
| Flan-T5 1-shot    | 0.13        | 3.91         | 0.02        | 0.96        | 1.66         | 0.05        | 1.12        |
| Flan-T5 2-shot    | 0.43        | 9.17         | 0.04        | <b>1.48</b> | <b>2.89</b>  | 0.18        | 2.37        |
| Gen-PINT 0-shot   | 0.22        | 3.03         | 0.76        | 0.25        | 0.59         | 0.61        | 0.91        |
| Gen-PINT 1-shot   | 0.26        | 10.16        | 0.70        | 0.75        | 0.77         | 0.64        | 2.21        |
| Gen-PINT 2-shot   | 0.28        | <b>10.21</b> | 0.79        | 0.87        | 0.73         | 0.67        | 2.26        |
| IGDM 0-shot       | 0.50        | 5.04         | 0.22        | 0.82        | 1.07         | 0.17        | 1.30        |
| IGDM 1-shot       | 1.07        | <b>10.20</b> | 0.73        | 1.05        | 1.82         | 0.46        | <b>2.56</b> |
| IGDM 2-shot       | <b>1.93</b> | <b>11.56</b> | 1.34        | 1.37        | 2.56         | <b>1.08</b> | <b>3.31</b> |

## V. MORE GENERAL MULTI-LABEL DATASETS

To investigate whether more general multi-label training data can further enhance the model performance, we used BlendClinic150, which covers 150 intents and 10 domains, as an alternative to the original BlendATIS for general training. The experimental results are shown in Tables V and VI. All methods improved in average accuracy and F1-score: For example, for IGDM, the 2-shot average accuracy increased from 23.42% to 27.15%, and the average F1-score increased from 2.01% to 3.31%. This indicates that larger-scale and more domain-rich training data can enhance the model’s generalization ability for unseen multi-label combinations, verifying the effectiveness of general data augmentation.

However, the performance improvement is not uniform. On datasets such as BlendSNIPS, the accuracy of IGDM decreased (for example, 2-shot from 26.26% to 23.32%). The reason is that more diverse intent categories can lead to an exponential increase in the number of multi-label combinations, and the frequency of each specific combination in the training data becomes extremely sparse. When the combination patterns of the target domain (such as personal voice assistants) do not match the training distribution, the model has difficulty generating the target-required label combinations precisely, and instead, it generates biases due to over-adaptation to sparse combinations. This shows that in the multi-label small-sample scenario, there is a trade-off between the comprehensiveness of the training data and the matching degree of the target domain’s combination.

## VI. HYPERPARAMETER ANALYSIS

To thoroughly investigate the practical impact of hyperparameters in the inner distribution maximization operation, we conducted a comprehensive sensitivity analysis on the maximum optimization steps  $T$  and the inner learning rate  $\eta$  using the RedWood dataset under three typical settings: 0-shot, 1-shot, and 2-shot. Specifically, we varied  $T$  from 1 to 6 and  $\eta$  from 20 to 120, using accuracy and F1-score as evaluation metrics. As shown in Fig. 5, the model demonstrates exceptional stability across all evaluation settings, with performance fluctuations caused by different hyperparameter combinations kept within a very narrow range. Particularly in the 1-shot scenario, when using the standard configuration of our method ( $T = 5$ ,  $\eta = 80$ ), the model achieved an accuracy of 94.57%. Upon further examination of the F1-score, this configuration also performed outstandingly with minimal fluctuations, maintaining a high level of performance overall. Across all hyperparameter combinations, the lowest accuracy remained at 93.92%, indicating that even with significant variations in the number of inner optimization steps and the learning rate within a certain range, the model’s performance did not experience a drastic drop. This directly validates the theoretical inference that, under the Lipschitz continuity assumption, perturbations are naturally constrained within the semantic preservation neighborhood. Given the dominant advantage in accuracy and the robustness of the hyperparameters, we selected  $T = 5$  and  $\eta = 80$  as the unified configuration for all experiments. This setting ensures efficient convergence while maximizing

the regularization effect of distribution maximization on the decision boundary.

## VII. STANDARD DEVIATIONS OF TABLES VI AND VII FOR THE MAIN ARTICLE

To verify the stability of the experiment, we conducted a single-label few-shot intention recognition experiment under five random seeds. Tables VII and VIII show the standard deviations of the accuracy and F1-scores for each method. The results indicate that the fluctuations of all methods were controlled within a relatively low range, proving that the experimental conclusion was not caused by randomness. Among these stable results, IGDM demonstrated highly competitive variance performance. Especially, in the few-shot setting, the 1-shot average standard deviation (accuracy) of IGDM was 1.00% and the 2-shot average was 0.97%, which was superior to or close to other adversarial training variants. This indicates that our method not only has stronger performance but also performs more robustly under different initialization conditions. This strong robustness is attributed to the design of the inner optimization of IGDM that fixes the instructions and only perturbs the query embeddings, avoiding the introduction of excessive randomness. Even in the face of different random initializations, IGDM can still learn consistent and highly generalizable decision boundaries. Therefore, the standard deviation analysis further confirms that IGDM significantly improves performance while maintaining good training stability.

## VIII. CONCLUSION

This supplementary material first provides a detailed theoretical analysis of Proposition IV.2. From the perspectives of gradient variance and information theory, it demonstrates that optimizing distribution maximization only on query samples is more effective than optimizing the instruction part simultaneously, avoiding the introduction of excessive randomness and the destruction of task semantics. Second, the material details the sources, domains, and scales of the 20 single-label and 7 multi-label intent recognition datasets involved in the experiments, ensuring the comprehensiveness and reproducibility of the experiments. Experimental results show that the proposed IGDM method significantly outperforms existing adversarial training methods (e.g., FGM, PGD, FreeLB) on both single-label and multi-label few-shot intent recognition tasks. In particular, under the 1-shot and 2-shot settings, IGDM achieves the highest accuracy and F1-score on most datasets, demonstrating superior generalization ability and robustness. In addition, the material discusses the preservation of sample semantics during distribution maximization, finding that perturbed samples maintain extremely high semantic similarity with the original samples. This indicates that IGDM improves model performance by enlarging the decision boundary rather than altering semantics. Finally, the value of data diversity is further verified by introducing more general multi-label training data (BlendClinic150), while the challenges caused by the sparsity of multi-label combinations are also pointed out.



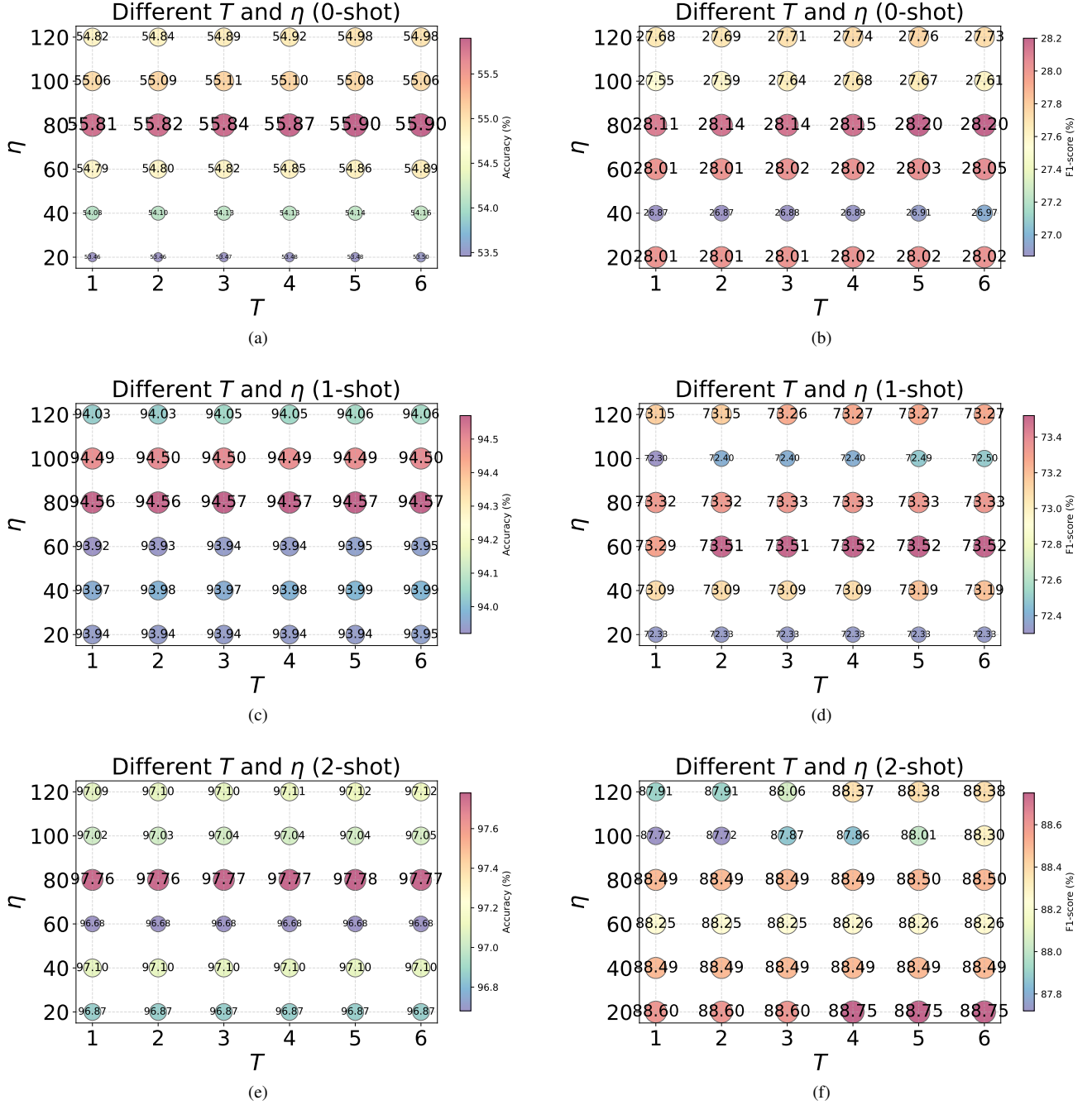


Fig. 5: Hyperparameter analysis on the RedWood dataset. (a), (c), (e) represent the results of accuracy (%), while (b), (d), (f) represent the results of F1-score (%). The hyperparameters we adopted in the experiment were set as  $T = 5$  and  $\eta = 80$ . Overall, this setting is the best. Although it sometimes falls short of the optimal level in terms of F1-score, its accuracy rate has a significant advantage in comparison.

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TABLE VII: Standard deviation (%) of Table VI in the main article.

| Method           | ATIS    | Bank77 | Clinic150 | Facebook | Hwu64  | LEYZER | Liu57 | MTOP |
|------------------|---------|--------|-----------|----------|--------|--------|-------|------|
| IG-FGM 0-shot    | 0.51    | 1.27   | 0.73      | 0.25     | 1.70   | 0.53   | 0.09  | 0.49 |
| IG-PGD 0-shot    | 0.83    | 0.65   | 0.68      | 0.16     | 1.72   | 1.60   | 0.27  | 0.40 |
| IG-FreeLB 0-shot | 4.59    | 4.66   | 1.99      | 4.36     | 8.12   | 2.69   | 3.90  | 0.58 |
| IGDM 0-shot      | 1.18    | 2.85   | 2.97      | 0.42     | 3.10   | 2.91   | 0.50  | 1.57 |
| IG-FGM 1-shot    | 0.56    | 1.22   | 0.26      | 0.25     | 1.04   | 0.85   | 0.33  | 0.16 |
| IG-PGD 1-shot    | 3.07    | 1.70   | 0.81      | 0.96     | 1.89   | 1.90   | 1.95  | 1.47 |
| IG-FreeLB 1-shot | 4.38    | 10.79  | 2.68      | 2.81     | 6.60   | 1.28   | 2.25  | 1.89 |
| IGDM 1-shot      | 1.32    | 1.20   | 0.36      | 0.46     | 1.62   | 1.10   | 1.03  | 0.67 |
| IG-FGM 2-shot    | 0.35    | 1.30   | 0.37      | 0.22     | 1.99   | 0.77   | 0.19  | 0.31 |
| IG-PGD 2-shot    | 3.42    | 1.35   | 0.68      | 0.66     | 1.78   | 1.43   | 2.27  | 1.43 |
| IG-FreeLB 2-shot | 4.63    | 7.70   | 1.84      | 3.78     | 7.23   | 5.76   | 2.44  | 0.47 |
| IGDM 2-shot      | 2.45    | 0.78   | 0.32      | 0.40     | 1.13   | 0.68   | 1.49  | 0.70 |
| Method           | RedWood | SLURP  | SNIPS     | StackE   | StackO | TOP    | TOPv2 | Avg. |
| IG-FGM 0-shot    | 1.02    | 1.24   | 0.44      | 2.94     | 0.10   | 0.12   | 1.58  | 0.87 |
| IG-PGD 0-shot    | 1.05    | 1.05   | 0.48      | 1.91     | 0.30   | 0.22   | 0.83  | 0.81 |
| IG-FreeLB 0-shot | 4.42    | 6.95   | 0.55      | 5.57     | 2.43   | 5.54   | 7.09  | 4.23 |
| IGDM 0-shot      | 3.30    | 3.01   | 0.33      | 1.90     | 0.64   | 0.31   | 3.47  | 1.90 |
| IG-FGM 1-shot    | 0.15    | 0.64   | 0.62      | 0.97     | 0.34   | 0.42   | 1.00  | 0.59 |
| IG-PGD 1-shot    | 1.25    | 2.13   | 1.10      | 1.62     | 1.92   | 1.67   | 2.23  | 1.71 |
| IG-FreeLB 1-shot | 1.18    | 3.59   | 0.95      | 4.56     | 3.16   | 3.98   | 10.80 | 4.06 |
| IGDM 1-shot      | 0.71    | 1.44   | 1.01      | 0.75     | 1.11   | 0.79   | 1.37  | 1.00 |
| IG-FGM 2-shot    | 0.81    | 1.06   | 0.72      | 2.46     | 0.15   | 0.46   | 1.54  | 0.85 |
| IG-PGD 2-shot    | 1.04    | 2.18   | 1.44      | 0.72     | 1.49   | 1.47   | 2.03  | 1.56 |
| IG-FreeLB 2-shot | 1.45    | 3.89   | 1.30      | 5.94     | 3.69   | 5.23   | 7.66  | 4.20 |
| IGDM 2-shot      | 0.58    | 1.20   | 1.06      | 0.96     | 0.92   | 0.72   | 1.19  | 0.97 |

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TABLE VIII: Standard deviation (%) of Table VII in the main article.

| Method           | ATIS    | Bank77 | Clinic150 | Facebook | Hwu64  | LEYZER | Liu57 | MTOP  |
|------------------|---------|--------|-----------|----------|--------|--------|-------|-------|
| IG-FGM 0-shot    | 1.09    | 1.49   | 1.54      | 2.21     | 1.09   | 0.75   | 1.70  | 0.99  |
| IG-PGD 0-shot    | 1.50    | 1.80   | 0.98      | 3.44     | 1.70   | 2.01   | 2.18  | 1.41  |
| IG-FreeLB 0-shot | 5.50    | 7.05   | 6.93      | 2.14     | 7.88   | 3.48   | 6.29  | 6.45  |
| IGDM 0-shot      | 1.03    | 2.35   | 1.93      | 4.60     | 1.26   | 2.54   | 1.93  | 1.79  |
| IG-FGM 1-shot    | 1.72    | 2.54   | 1.57      | 3.21     | 1.03   | 1.07   | 2.12  | 2.83  |
| IG-PGD 1-shot    | 2.14    | 3.51   | 2.59      | 9.51     | 4.58   | 3.04   | 4.72  | 3.14  |
| IG-FreeLB 1-shot | 4.31    | 15.83  | 6.99      | 4.39     | 8.01   | 2.98   | 8.76  | 10.70 |
| IGDM 1-shot      | 1.22    | 4.19   | 2.85      | 5.87     | 4.40   | 1.36   | 7.03  | 4.87  |
| IG-FGM 2-shot    | 1.55    | 2.69   | 1.88      | 2.11     | 1.96   | 2.12   | 2.62  | 2.08  |
| IG-PGD 2-shot    | 3.11    | 3.07   | 1.80      | 8.02     | 6.38   | 3.04   | 3.36  | 3.24  |
| IG-FreeLB 2-shot | 6.27    | 14.22  | 7.63      | 3.25     | 8.60   | 1.58   | 9.12  | 8.42  |
| IGDM 2-shot      | 1.98    | 2.23   | 2.91      | 5.13     | 3.37   | 2.54   | 4.38  | 2.83  |
| Method           | RedWood | SLURP  | SNIPS     | StackE   | StackO | TOP    | TOPv2 | Avg.  |
| IG-FGM 0-shot    | 0.94    | 0.84   | 0.56      | 1.59     | 1.35   | 2.50   | 1.27  | 1.33  |
| IG-PGD 0-shot    | 1.10    | 1.09   | 2.63      | 2.21     | 4.15   | 2.39   | 0.62  | 1.95  |
| IG-FreeLB 0-shot | 5.60    | 7.28   | 0.60      | 5.49     | 6.03   | 1.84   | 9.00  | 5.44  |
| IGDM 0-shot      | 1.84    | 0.97   | 3.10      | 4.68     | 4.89   | 3.40   | 0.71  | 2.47  |
| IG-FGM 1-shot    | 1.26    | 0.89   | 0.69      | 0.79     | 1.46   | 0.55   | 1.87  | 1.57  |
| IG-PGD 1-shot    | 2.56    | 5.33   | 3.02      | 1.37     | 7.74   | 2.40   | 6.58  | 4.15  |
| IG-FreeLB 1-shot | 5.43    | 6.76   | 1.83      | 2.33     | 10.39  | 4.14   | 17.02 | 7.32  |
| IGDM 1-shot      | 2.58    | 3.19   | 4.28      | 0.97     | 5.27   | 5.82   | 1.49  | 3.69  |
| IG-FGM 2-shot    | 1.90    | 0.95   | 0.83      | 1.94     | 1.37   | 1.96   | 2.95  | 1.93  |
| IG-PGD 2-shot    | 2.48    | 4.56   | 3.09      | 1.06     | 4.74   | 5.51   | 7.85  | 4.09  |
| IG-FreeLB 2-shot | 5.47    | 7.15   | 2.19      | 2.56     | 7.83   | 2.00   | 14.51 | 6.72  |
| IGDM 2-shot      | 2.39    | 2.78   | 2.59      | 1.27     | 5.02   | 4.77   | 1.64  | 3.06  |

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